# Trigonometric Equation

Mind map

## Types of solutions

- Principal: $[0, 2\pi)$
- General: all, with $n \in I$
- Particular: given interval

## Standard results

- $\sin x = 0 \Rightarrow n\pi$
- $\cos x = 0 \Rightarrow (2n+1)\frac{\pi}{2}$
- $\sin x = \pm 1,\ \cos x = \pm 1$

## General formulas

- $n\pi + (-1)^n \alpha$
- $2n\pi \pm \alpha$
- $n\pi + \alpha$
- Squares: $n\pi \pm \alpha$

## Solutions in an interval

- Put $n = 0, \pm 1, \ldots$
- Count on the graph
- Equal forms: compare in $[0, 2\pi)$

## Type 1: factorise

- Quadratic in one ratio
- Do not cancel factors
- Keep values in $[-1, 1]$

## Type 2: $a \sin x + b \cos x = c$

- Divide by $\sqrt{a^2 + b^2}$
- $|c| \leq \sqrt{a^2 + b^2}$
- Homogeneous: divide by $\cos^2 x$

## Type 3: transform

- $\sin mx = \sin nx$: $n$ even/odd
- $C \pm D$ to factorise
- Product to sum

## Type 4: substitute

- $t = \sin x \pm \cos x$
- $t = \sin 2x,\ \tan \frac{x}{2},\ a^{\sin^2 x}$
- $t$ within its range

## Type 5: boundedness

- max of one side = min of other
- $\sin^m x + \cos^n x \leq 1$
- Common solution of conditions

## Type 6: squaring

- Extraneous roots appear
- Verify in original equation
- Valid values $+2n\pi$

## Inequalities

- One period, then add period
- Read from the graph

## Systems

- Common values in $[0, 2\pi)$
- $x + y$ and $x - y$
- Different $n, m$